

Light and Photosynthesis

Student worksheet | Biology 2e, Chapter 8 | 45-60 minutes

Before you begin: prepare for the investigation

What is infiltration?

Infiltration is the movement of liquid into spaces within a material. When those spaces contain trapped air, reducing the pressure can help remove the air so liquid can take its place.

Place a porous material in liquid inside a needle-free syringe. Gently expel excess air, cover the tip, and pull back the plunger to lower the pressure. Trapped gas can expand and escape. Uncover the tip and gently release the vacuum; liquid can enter the spaces as normal pressure returns. Repeated cycles may be needed.

Before testing, predict how this preparation might affect your sample. Record your prediction and reasoning on the next page.

Run a controlled comparison

The virtual lab guides you through three dark trials, three low-light trials, and three bright-light trials. For each trial:

1. In Collect observations, use Pull plunger and Release vacuum three times to prepare the disks.
2. Click Add disks to cup. The cup stays empty until you transfer the prepared disks.
3. Check the assigned lamp setting: Off (0%), Low (25%), or High (100%). Select it if needed. If the correct setting is already highlighted, you can start without clicking it again. The setting stays fixed during the trial.
4. Click Start trial and observe for 20 simulated minutes. You can pause, change timer speed, advance two minutes, or run the remaining time. The app records readings every two simulated minutes; copy them into your observation table.
5. Review the readings, then click Collect data and continue to save the results and advance. Repeat with fresh disks until all nine trials are collected.

After the controlled comparison, use Explore other settings to investigate freely. The exploration activity is at the end of this lab worksheet; your comparison data stay saved separately.

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Name: _____ Class: _____ Date: _____

1. Predict and plan

How does light level affect the time it takes leaf disks to float?

Predict how infiltration and light conditions might affect the leaf disks. Complete the guided comparison using the assigned light level for each 20-minute trial. After collecting all nine trials, explore other settings and compare your observations with your prediction.

Prediction and scientific reasoning:

Variables and controls, or sampling plan:

Keep consistent: 10 same-sized disks, same leaf type, 0.2% sodium bicarbonate solution with a trace of detergent, solution depth, and temperature. Only light level changes.

Select your route

Part	Virtual / Classroom	Equipment or conditions
Dark control		
Low light		
Bright light		

Use the same questions for either route. Always label the source of each dataset.

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2. Collect observations

Record floating disks (of 10). Repeat each treatment with fresh material for three independent trials.

Treatment / trial	0 min	2 min	4 min	6 min	8 min	10 min	12 min	14 min	16 min	18 min	20 min
Dark control / 1											
Dark control / 2											
Dark control / 3											
Low light / 1											
Low light / 2											
Low light / 3											
Bright light / 1											
Bright light / 2											
Bright light / 3											

Source (virtual/classroom) for each treatment: _____

Trial summaries

Treatment	Trial 1	Trial 2	Trial 3	Mean
Dark control				
Low light				
Bright light				

Metric: ET50 in minutes. Use the first crossing of five disks; interpolate between readings. If not reached by 20 min, report that limit. Calculate a mean only when all three trials reach ET50.

Observations and graph

Graph time on the x-axis and the recorded measurement on the y-axis on separate graph paper, or use the app. Label treatments, units, and data sources. Record unusual observations below.

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3. Explain the evidence

ET50 is the interpolated time when 5 of 10 disks float. Lower ET50 suggests faster net oxygen accumulation under matched conditions. If five disks never float, report "not reached within 20 minutes"; do not treat that as a zero rate.

1. Which light condition had the shortest ET50? Support your claim with values from at least two treatments and discuss variation between trials.

2. Why are the dark control and matched leaf preparation necessary?

3. Explain how oxygen accumulation makes disks float and why this is an indirect measure of net activity, not a direct measurement of sugar production.

4. Identify one limitation or source of uncertainty and propose an improved investigation.

Before sharing: check units, source labels, calculations, and evidence. The app saves on this device only; download or print your report.

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4. Explore after the controlled comparison

After collecting all nine trials, choose Explore other settings. Prepare a fresh set of disks in the syringe, then click Add disks to cup. Before testing a change, predict what you expect to observe.

Change the lamp during a run, or choose any intensity from 0 to 100%. After 20 minutes, resume to observe up to 120 minutes. Try high light, darkness, then high light again. Turning off the lamp does not pause time.

Exploration is separate from your nine comparison trials and does not change their results. Download your data before restarting the exploration.

My prediction and reasoning:

My change (setting and simulated time):

What happened immediately? What happened later? Compare your observations with your prediction and propose an explanation.

CSV and text reports include your exploration observations. Model rates are illustrative, not predictions for real leaves.

Cut & sort: matter and energy

Optional introduction or extension | 10-15 minutes | Biology 2e, chapters 7 and 8.

Choose Card Activity on the lab overview to open the separate circular photosynthesis and respiration model, or arrange the printed cards on chart paper. Use two molecule sets for both halves at once, or reuse one 19-card set as you switch processes.

Print cutout pages in color, single-sided, at 100% scale. Cut along dashed lines. Use scissors safely. Group work: one person arranges, one counts atoms, and one explains; then rotate roles.

Model photosynthesis. Sort all the molecule cards into reactants and products. Add plus signs and the arrow, and decide where light and chlorophyll belong.

Count carbon, hydrogen, and oxygen atoms separately on each side. Explain how your arrangement shows conservation of matter and the transfer of energy. Energy symbols are annotations, not part of this simplified atom inventory.

This is an overall net equation, not a sequence of reaction stages. The glucose diagram shows an atom inventory, not a molecular structure.

Name: _____ Class: _____ Date: _____

My equation and symbol placement:

Atom counts: Reactants C ____ H ____ O ____ | Products C ____ H ____ O ____

Explain where energy enters, is stored, or leaves this process:

PHOTOSYNTHESIS • CUT & SORT

Cut on dashed lines. Keep all three sheets together.

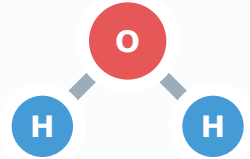
Build a balanced equation. Place light and chlorophyll where they help.



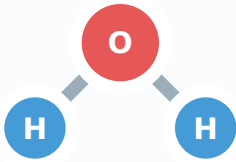
Oxygen



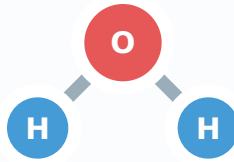
Carbon dioxide



Water



Water



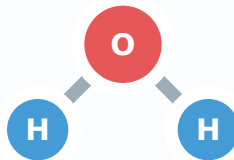
Water



Oxygen



Carbon dioxide



Water



Carbon dioxide

Atom colors: C = gray • O = red • H = blue

Student cutouts 1/3 • Print in color, single-sided, at 100%.

PHOTOSYNTHESIS • CUT & SORT

Cut on dashed lines. Keep all three sheets together.

Build a balanced equation. Place light and chlorophyll where they help.

+



Combine with

O₂



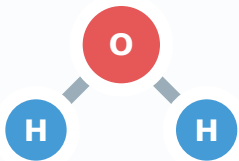
Oxygen

CO₂



Carbon dioxide

H₂O



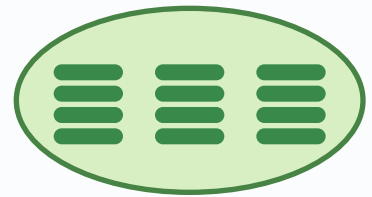
Water

+



Combine with

CHLOROPHYLL



Green pigment in chloroplasts
Simplified chloroplast diagram
Light-absorbing pigment

C₆H₁₂O₆



Atom inventory (not structure)

Glucose

O₂



Oxygen

LIGHT ENERGY



Sunlight

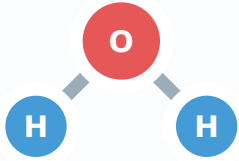
Atom colors: C = gray • O = red • H = blue

Student cutouts 2/3 • Print in color, single-sided, at 100%.

PHOTOSYNTHESIS • CUT & SORT

Cut on dashed lines. Keep all three sheets together.

Build a balanced equation. Place light and chlorophyll where they help.



Water



Reaction arrow



Oxygen



Oxygen



Carbon dioxide



Carbon dioxide

Atom colors: C = gray • O = red • H = blue

Student cutouts 3/3 • Print in color, single-sided, at 100%.